

Mrunal Kotkar

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Professional Summary

Graduate Computer Science student with 3+ years of professional experience building scalable, distributed, and cloud-native systems. Proficient in Java, Python, and C++ with hands-on expertise in Spring Boot, AWS, microservices, and CI/CD pipelines. Deeply passionate about AI/ML, from building production LLM-powered agentic systems to deploying deep learning models. Proven track record of owning ambiguous, high-impact projects end-to-end in fast-paced, collaborative engineering environments.

Education

San Jose State University	(Expected) May 2027
M.S. in Computer Science	GPA 4.00/4.00
College of Engineering, Pune	May 2022
B.Tech. in Computer Engineering	GPA 3.93/4.00

Work Experience

UBS Senior Software Engineer	Dec 2023 - July 2025
- Independently owned and drove end-to-end delivery of a multi-threaded risk analytics services in Spring Boot, improving system throughput by 30% using REST APIs and asynchronous data pipelines. - Architected distributed approval path modules handling 1,000+ transactions/min across multi-region (APAC, EMEA) clusters with Kubernetes orchestration. - Integrated Elasticsearch and GitLab CI/CD, achieving 25% faster query performance and seamless versioned deployments on Linux. Collaborated in Agile development to deliver secure, scalable microservices leveraging Dockerized environments.	
Credit Suisse Technology Analyst	June 2022 - Dec 2023
- Built Grafana dashboards and Python alert systems, boosting observability and cutting manual work by 30% - Optimized Django/C++ backend services, decreasing system latency by 15%	

Projects

Image Generation using VAE, GAN and Diffusion Models Machine Learning, Pytorch, KID & FID
Benchmarked VAE, GAN, and Diffusion models, evaluating 1,000+ generated images to quantify differences in quality, stability, and reconstruction accuracy
Smartphone-Based Fall Detection Python, TensorFlow/Keras, scikit-learn, Pandas, NumPy
Built an ML pipeline on MobiFall v2.0 dataset with 100 Hz resampling, and multi-size windowing. Compared Deep learning models for fall detection and activity classification. Achieved 99.35% accuracy with a 1D CNN.
Multi-Agent Code Review & PR Assistant Python, LangGraph, LangChain, GitHub API
Built a multi-agent PR review system using LangGraph, with specialized agents for security, quality, and summarization collaborating via tool-calling and shared memory to automate code reviews end-to-end.
Thanos - AWS Misconfig Detection System Python, Terraform, AWS Resources, Typescript
Developed an automated AWS security scanner detecting misconfigurations across multiple resources, reducing manual audit time by 70%. Integrated MCP server and AI agent workflows to automate cloud security remediation recommendations.

Technical Skills

Programming Languages: Java, Python, C, C++, TypeScript, JavaScript, Bash, HTML/CSS, SQL
Frameworks & Systems: Spring Boot, Django, NodeJS, React, Linux, UNIX, Networking (TCP/IP, Sockets, Process Scheduling), Docker, Kubernetes, Ansible, CI/CD (GitLab, Jenkins), Cloud Infrastructure, REST APIs, Containerization, Open Source
AI & Agentic Systems: LLMs, RAG, AI Agents, LangChain, LangGraph, TensorFlow, PyTorch, CUDA, Generative AI, GPU Acceleration, Vector Databases, Model Context Protocol

Leadership & Achievements

Winner - AWS Cloudathon 2026: 2nd Place, Solved real-world cloud challenges that engineers face daily at their workplace, competing against 300+ peers in a hackathon organized by AWS at San Jose State University.
Winner - Ai Tinkerers Hackathon 2025: Built an AI-powered Sprint Planning Assistant leveraging LLMs and agentic frameworks (Composio, Redis) to automate Jira ticket creation. Demonstrated applied AI engineering and end-to-end product ownership.
Recipient - SJSU Computer Science Department Scholarship: In recognition of excellent merit, academic achievement, leadership, and community involvement (2026-2027)

Publications

Reflection based Distributed Denial of Service Attack Detection System	January 2023
<u>Land Registration System Using Blockchain</u>	June 2024